

From Waste to Material: Aloe Vera Rind as a Source of Cellulose Nanocrystals

An Environmentally Responsible Approach to Agricultural Waste Valorization

Current Practice & Environmental Effects

Aloe barbadensis Miller is a plant cultivated for its medicinal gel. The global market is growing rapidly due to surging demand for natural ingredients in personal care, increased health consciousness, and for its versatility.

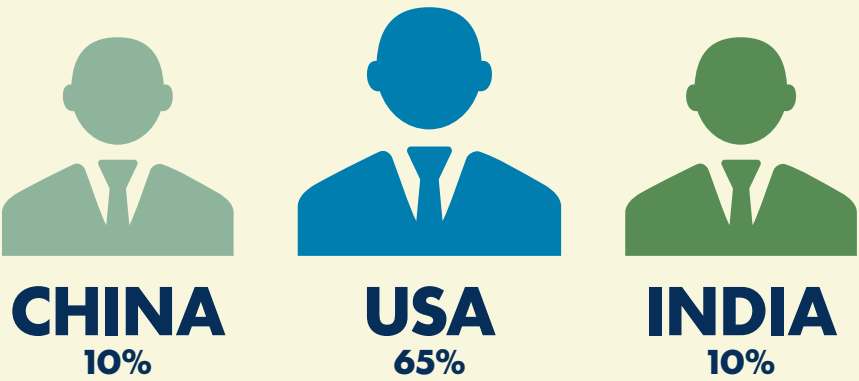
The Problem

A single aloe vera gel processing facility generates up to 4,000 kg of waste per month where 20-30% of the plant's weight is leaf. The leaf or rind is considered agricultural waste and current disposal methods are: landfill or low-value fertilizer with no recovery of material value.

Gel Recovery by Extraction Method

43.57%	Mechanical Filleting
57.58%	Hand Filleting
70.28%	Whole Leaf Processing (WLP)
98.13%	WLP Through Mechanical Filleting

Primary Markets



Environmental Harms
Air Pollution

Landfill Disposal

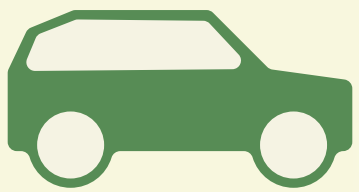
Soil Contamination

Groundwater Leaching



Aloe Vera Market
Projection at 8.6% CAGR
**\$2.65
BILLION**
by 2027

Landfill Emissions
0.41 MTCO₂e
per wet
tonne
=
~4500



The ERA: Cellulose Nanocrystal Recovery from Aloe Vera Rinds

Aloe vera rind contains 42-58% cellulose by dry weight which is sufficient to extract cellulose nanocrystals (CNCs), diverting waste from landfill and replacing virgin cellulose demand.

Extraction method is proven

Garcia-Zaleta et al. (2025): CNC successfully extracted from aloe vera rind via sulfuric and citric acid hydrolysis — crystallinity confirmed at 63–64%

Vasilopoulos et al. (2024): Microwave-assisted extraction + sulfuric acid hydrolysis optimized for energy efficiency

Why Aloe Vera Rinds?

Low Lignin Content
19.12%

High Cellulose Content
53.96%

Clear Value Added Compared to Landfill or Fertilizer

Landfill Disposal

Eliminates 0.19–0.41 MTCO₂e/wet tonne

What it Replaces

Virgin Wood Pulp Cellulose

Avoids forest harvesting, kraft pulping, and bleaching

Petroleum-Based Synthetic Nanomaterials

Replaced by a biodegradable alternative

Circular Economy Connection waste from gel extraction becomes feedstock for CNC production which closes the biological material loop as defined by Cradle-to-Cradle principles.

Why Keep the Current Practice?

Extraction of CNC from aloe vera is not yet commercially viable. The standard chemistry involved in CNC extraction are hazardous carcinogenic acids which require energy-intensive processes, generating high-salinity wastewater which drives production costs as high as \$25/kg of CNC. Even if greener and safer alternatives exist, due to the low quantity of waste indicated by small producers at 1 tonne per month, it is not economically justifiable to have processing infrastructure per facility.

The market for CNC makes entry difficult due to large players like CelluFroce Inc. Which produce CNC at an industrial scale at 1 tonne per day from wood pulp, with decades of process and supply chain refinement - aloe vera CNC will be commercially marginal in comparison. Moreover, the end market for CNC packaging is emerging and valued at only \$20.3 million in 2024 with food-contact regulatory approvals still pending. Building out infrastructure is a real financial risk before demand exists.

The rind already has a use and using it as a fertilizer closes a natural nutrient loop without added chemicals, no capital equipment and no market risk.

Counter-Argument

The issue with the current practice is that it rests on solvable problems, not fundamental barriers. Deep eutectic solvents (DES) which are non-toxic, low-cost, and achieve 68-78% CNC yield, removing the safety and waste concerns concerning conventional sulfuric acid hydrolysis. The scale argument dissolves under a regional biorefinery model, which is standard in agricultural processing, which aggregates rind from multiple producers to justify shared infrastructure.

The market for aloe vera-derived CNC is differentiated from commodity CNC. This is a waste-derived, traceable, bio-compatible CNC suited to cosmetics, pharmaceuticals, and speciality packaging, particularly for aloe vera brands with active sustainability commitments. Aloe vera derived CNC is for a distinct market, and not a direct competitor with CelluForce.

The energy and emissions concern also shifts when viewed through a full lifecycle lens: producing 1 kg of CNC generates approximately 11.39 kgCO₂e which is favourable compared to synthetic nanomaterials and petroleum-based packaging components. While the rind-as-fertilizer argument sounds clean, at volume it generates odour pollution, runoff, and leachate - and CNC extraction doesn't eliminate composting, it precedes it.

Finally, pre-commercial markets require early infrastructure investment by definition. CNC demand and aloe vera processing growth are parallel trends, making now the strategically rational movement to establish the supply chain - not after competitors do.



Conclusion

Aloe vera gel processing generates up to 4,000 kg of rind waste per facility per month. That rind contains 42–58% cellulose by dry weight and produces 0.19–0.41 MTCO₂e per wet tonne when landfilled. CNC extraction from this waste stream is technically proven, the output displaces virgin and petroleum-based materials, and the process chemistry is becoming cleaner. Diverting an existing waste stream into a high-value biodegradable material is better than landfilling it.

3 Caveats

1. Conventional sulfuric acid extraction generates ~11.39 kgCO₂e per kg of CNC — the net gain only materialises when the CNC replaces something more emissions-intensive, and when greener DES chemistry is used.
2. The CNC packaging market is still pre-commercial at \$20.3M in 2024, meaning this ERA is built ahead of confirmed mass-market demand.
3. And the fertilizer pathway, while lower-value, remains a valid option for the smallest operators — it should be retained as a complementary final step, not discarded.

The ERA applies most cleanly at commercial-scale facilities, using DES chemistry, aggregating rind from multiple producers through a regional model. Under those conditions, the environmental case is solid, the waste stream is real and growing, and the material produced has documented value across multiple industries.

Garg, S., & Avanthi, A. (2025, October 21). Overcoming cost, energy, and process barriers for industrially viable nanocellulose production - discover Applied Sciences. SpringerLink. <https://link.springer.com/article/10.1007/s42452-025-07650-6>

Khalidoun, K., Fdil, N., & Ali, M. A. (2024, February 6). Exploring Aloe vera: A comprehensive review on extraction, chemical composition, biological effects, and its utilization in the synthesis of metallic nanoparticles. ScienceDirect. <https://www.sciencedirect.com/science/article/abs/pii/S1878818124000355#article>

Koçer, A. T., Erarslan, A., & Balkanlı, D. (2023, August 28). Pyrolysis of Aloe vera leaf wastes for biochar production: Kinetics and thermodynamics analysis. ScienceDirect. <https://www.sciencedirect.com/science/article/abs/pii/S0926669023011196>

Majeed, H., Iftikhar, T., & Manzoor, R. (2024, September 6). Extraction and characterization of novel alternative cellulosic fiber for sustainable textiles from Aloe barbadensis Miller stems (agricultural waste). ScienceDirect. <https://www.sciencedirect.com/science/article/pii/S2405844024134597>

MENAFN. (2023, October 30). Global Aloe Vera Extract Market Expected To Reach \$3.65 Billion By 2027, Demonstrating A CAGR Of 8.6%. Nexis Uni. <https://advance.lexis.com.proxy.library.carleton.ca/document?cid=a7125f92-3ef1-4ac2-8a33-0ac93e256500&pdocrfullpath=%2Fshared%2Fdocument%2Fnews%2Fnews%3AcontentItem%3A6C16-T6K1-JBR8-B4B1-00000-00&pdsourcegroupingtype=&pdaleriemail=False&pdcontentcomponentid=473845&pdmfid=1519360&pdurlapi=true>

Pulidindi, K. (2025, October). Cellulose nanocrystals for packaging applications market size. Global Market Insights Inc. <https://www.gminsights.com/industry-analysis/cellulose-nanocrystals-for-packaging-applications-market>

Semerel, J., John, N., Dehaen, W., & Fardim, P. (2022, October 27). Valorization of Aloe barbadensis Miller. (Aloe vera) Processing Waste. ScienceDirect. <https://www.sciencedirect.com/org/science/article/pii/S2164637522001913>

Vavrková, M. D. (2019, October 3). Landfill impacts on the environment-review. MDPI. <https://www.mdpi.com/2076-3263/9/10/431>

Zhang, L., Jia, X., Ai, Y., Huang, R., Qi, W., He, Z., Klemes, J. J., & Su, R. (2022, February 22). Greener production of cellulose nanocrystals: An optimised design and life cycle assessment. ScienceDirect. <https://www.sciencedirect.com/science/article/abs/pii/S0959652622007065>